MIL-I-50360(MU)
15 January 1971
SUPERSEDING
FA-PD-MI-2428, Rev. 2
26 April 1963

MILITARY SPECIFICATION
INITIATORS, CARTRIDGE ACTUATED DELAY, M51 AND M88
METAL PARTS ASSEMBLY

1. SCOPE

1.1 This specification covers Initiators, Cartridge Actuated Delay, M51 and M88, Metal Parts Assembly. This assembly will ultimately be loaded and used in conjunction with other propellant actuated devices in aircrew emergency escape system.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein:

SPECIFICATIONS
Military

MIL-A-2550	- Ammunition and special weapons, General Specification for

STANDARDS
Military

MIL-STD-105	- Sampling procedures and tables for Inspection by Attributes
MIL-STD-109	- Quality Assurance Terms and Definitions
MIL-STD-413	- Visual Inspection Guide for Rubber O-Rings

DRAWINGS
Munitions Command

D10520909	- Initiator Cartridge Actuated, Delay, Metal Parts Assembly
C8846787	- Container Ammunition, Fiber M458 for Initiators, Cartridge Actuated Delay
C8846788	- Box, Packing Ammunition for Initiators, Cartridge Actuated, Delay in Fiber Container M458
IEL 10520909	- Inspection Equipment List for Initiator, Cartridge Actuated, Delay, Metal Parts Assembly

(Copies of specifications, standards and drawings required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

FSC 1377

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2.2 Other publications - The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bid or request for proposal shall apply.

ASA B46.1 - Surface Texture

(Application for copies should be addressed to the American Standards Association, 10 East 40th Street, New York, N.Y. 10016).

3. REQUIREMENTS

3.1 General - Materials, parts and assemblies shall comply with the requirements specified on the applicable drawings and referenced specifications.

3.2 Mechanical properties - Mechanical properties required by the drawings shall be uniform throughout the part within specified tolerances. Requirement for all alloy steel parts shall be achieved by quenching and tempering.

3.3 Hydrostatic pressure - The initiator chamber and cap shall withstand a hydrostatic pressure of 14,000 to 15,000 PSI - pounds per square inch for 15 seconds, minimum, without deformation or mechanical failure (see 6.2).

3.4 Shear Pin Strength - The shear pin after application of protective finish, shall shear when a load of 41 pounds plus or minus 5 pounds is applied at the steady rate of 0.125 inch per minute.

3.5 Protective finish - Protective finish shall be in accordance with the applicable specifications and shall apply to all surfaces of the component regardless of configuration.

3.6 Residual magnetism - The metal parts assembly shall not deflect the indicator of the compass more than three degrees in either direction when passed in front of the compass at a distance of 6 inches.

3.7 Workmanship - The requirement for workmanship areas specified on the applicable drawings, referenced specifications and the following·

3.7.1 Metal defect - The part shall be free from cracks, splits, cold shuts, inclusions, porosity or other metal defects.

3.7.2 Burr - The part shall not have a burr which might interfere with the assembly of function of the item or be injurious to personnel handling the item.

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3.7.3 Foreign matter - The part or assembly shall not contain chips, dirt, grease, oil, rust, corrosion or other foreign matter.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Quality Assurance Terms and Definitions - Reference shall be made to Mil-Std-109 to define the quality assurance terms used.

4.1.2 Inspection - Inspection shall be in accordance with Mil-A-2550.

4.2 Initial production sample - At the beginning of regular production, a sample shall be submitted in accordance with contract requirements and shall consist of 50 units of each heat treated lot, for hardness test and the specified number of units for protective finish tests in accordance with the specification referenced on the drawing to be tested in accordance with Table I and 10 unassembled component for visual and dimensional examination.

TABLE I

Paragraphs				Type of Test	
Requirements	test	Test	Sample Size	Destr	Non-Destr
3.2	4.4.1	Hardness	50	X	-
3.3	4.4.2	Hydrostatic Pressure	Each	-	X
3.4	4.4.3	Shear pin strength	50	X	-
3.5	4.4.4	Protective Finish	As required by reference specification.	X	-
3.6	4.4.5	Residual magnetism	Each	-	X

The sample shall be manufactured using the same materials, equipment, processes and procedures as used in regular production. All parts and materials, including packaging and packing shall be the same as used for regular production and shall be obtained from the same source of supply.

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4.2.1.1 Examination and tests - After examination and provisional acceptance at source, the sample shall be inspected for all requirements of the drawings and specifications at a Government laboratory or such other facility specified in the contract.

4.2.1.2 Initial production failure - Failure of the sample to comply with the requirement of the drawings and specifications shall result in sample disapproval.

4.3 Inspection provisions

4.3.1 Lot

4.3.1.1 Submittal of product - The product shall be submitted in accordance with Mil-Std-105.

4.3.1.2 Size of Lot - A lot shall consist of an identifiable group of metal parts assemblies consisting of component produced by a homogeneous process.

4.3.2 Examination - One hundred percent examination shall be performed for all critical defects. Examination for major and minor defects shall be performed on a defect basis in accordance with the classification of defects contained therein, using the sampling plan and associated criteria in Tables II and III. For an individual defect (single characteristic) the criteria in table II shall be used. For total defect within a major or minor class, the criteria in table III shall be used. All non conforming material shall be rejected. The roughness comparison specimens prescribed in ASA B46.1 shall be used as a basis of comparison for surface roughness determination. Four power (4X) magnification may be used in performing visual examination for such defect as burr and foreign matter.

TABLE II

Lot Size	Sample Size	Major		Minor	
		Accept	Reject	Accept	Reject
1 - 300	100% examination	-	-	-	-
301 - 1300	110	1	2	2	3
1301 - 3200	150	2	3	3	4
3201 - 8000	225	3	4	4	5
8001 - 22000	300	4	5	5	6

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TABLE III

Sample Size for Major Defect Number of defects in class	Total Allowable Defect Per Sample Size 110	150	225	300
2 - 3	1	2	3	4
4 - 7	2	3	4	5
8 - 19	3	4	5	7
20 - 43	4	5	8	10
Over 43	6	8	11	14

Sample Size for Minor Defect Number of defects in class	Total Allowable Defect Per Sample Size 110	150	225	300
2 - 3	2	3	4	5
4 - 7	3	4	5	7
8 - 16	4	5	8	10
17 - 48	6	8	11	14
Over 48	8	11	17	20

4.3.2.1 Chamber, Initiator, Dwg D8597928 Covering a Detail of Drawing D1052O909

Defect	Method of Inspection
CRITICAL	
1. Metal defective	Visual
MAJOR	
101. Pitch diameter of large external thread, min	Gage
102. Major diameter of large external thread, min	Gage
103. Pitch diameter of small external thread, min	Gage
104. Major diameter of small external thread, min	Gage
105. Pitch diameter of internal thread, max	Gage
106. Minor diameter of internal thread, max	Gage
107. Concentricity at datum location with pitch diameter of small external thread	Gage
108. Concentricity of pitch diameter of large external thread with cavity	Gage
109. Concentricity of large external thread pilot with large external thread	Gage
110. Concentricity of counterbore with cavity including perpendicularity with base of counterbore	Gage

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4.3.2.1 Chamber, Initiator, Dwg D8587928 Covering a Detail of Drawing D10520909 - Cont'd

111.	Diameter of counterbore	Gage
112.	Diameter of large thread pilot	Gage
113.	Depth of cavity	Gage
114.	Length to datum location	Gage
115.	Thickness of wall, min	Gage
116.	Protective finish inadequate or defective	Visual
117.	Surface finish of cavity and angle of gas port improper	Visual
118.	Burr	Visual
119.	Foreign matter	Visual

MINOR

201.	Pitch diameter of large external thread, max	Gage
202.	Pitch diameter of small external thread, max	Gage
203.	Pitch diameter of internal thread, min	Gage
204.	Diameter of small external thread relief	Gage
205.	Diameter of small external thread pilot	Gage
206.	Diameter of hole in gas port	Gage
207.	Diameter of cavity	Gage
208.	Diameter of chamfer at internal thread	Gage
209.	Depth of counterbore	Gage
210.	Depth of threaded hole	Gage
211.	Length of gas port	Gage
212.	Length of small thread relief	Gage
213.	Length from face of gas port to small thread	Gage
214.	Length of large external thread pilot	Gage
215.	Total length	Gage
216.	Length of large external thread	Gage
217.	Thickness of wall	Gage
218.	Marking missing, incorrect or illegible	Visual
219.	Radius or chamfer missing or incorrect	Visual
220.	Surface finish improper, except as otherwise classified	Visual

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1.1 Scope. This specification covers the following:

4.3.2.2 Filter, Drawing C10520414, Covering a Detail of Dwg. D10520909

	Defect	Method of Inspection
CRITICAL		
1.	Metal defective	Visual
MAJOR		
101.	Pitch diameter of thread, min	Gage
102.	Major diameter of thread, min	Gage
103.	Concentricity of thread with center hole and outside diameter	Gage
104.	Symmetry of holes with outside diameter	Gage
105.	Diameter of center hole	Gage
106.	Diameter of holes	Gage
107.	Depth of center hole	Gage
108.	Depth of slot, max	Gage
109.	Location of holes	Gage
110.	Angular location of holes	Gage
111.	Nylon patch not properly located or fused	Manual and Visual
112.	Number of holes incorrect	Visual
113.	Protective finish inadequate or defective	Visual
114.	Burr	Visual
115.	Foreign matter	Visual
MINOR		
201.	Pitch diameter of thread, max	Gage
202.	Diameter of undercut	Gage
203.	Diameter of body	Gage
204.	Depth of slot, min	Gage
205.	Total length	Gage
206.	Length of body	Gage
207.	Width of undercut, max	Gage
208.	Width of slot	Gage
209.	Location of slot	Gage
210.	Marking missing, incorrect or illegible	Visual
211.	Radius or chamfer missing or incorrect	Visual
212.	Surface finish improper	Visual

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4.3.2.3 Cap, Initiator, Dwg. D10520473 Covering a Detail of Drawing D10520909

Defect	Method of Inspection
CRITICAL	
1. Metal defective	Visual
MAJOR	
101. Pitch diameter of large internal thread, max	Gage
102. Minor diameter of large internal thread, max	Gage
103. Pitch diameter of small internal thread, max	Gage
104. Minor diameter of small internal thread, max	Gage
105. Pitch diameter of external thread, min	Gage
106. Major diameter of external thread, min	Gage
107. Pitch diameter of pin hole thread, max	Gage
108. Minor diameter of pin hole thread, max	Gage
109. Concentricity of striker hole with small cavity	Gage
110. Concentricity of large internal thread undercut with large thread	Gage
111. Concentricity of cartridge head recess with large internal thread	Gage
112. Concentricity of large internal thread with small cavity	Gage
113. Concentricity of chamfer at striker hole with small cavity	Gage
114. Concentricity of small internal thread with small cavity	Gage
115. Symmetry of pin hole with small cavity	Gage
116. Symmetry of flats with face at small end	Gage
117. Diameter of small cavity	Gage
118. Diameter of large thread undercut	Gage
119. Diameter of boss, min	Gage
120. Diameter of striker hole, min	Gage
121. Diameter of pin hole, max	Gage
122. Diameter of cartridge head recess	Gage
123. Diameter of counterbore at small thread cavity	Gage
124. Diameter of mouth at small thread cavity	Gage
125. Depth of cartridge head recess	Gage
126. Depth to bottom of counterbore small threaded end, max	Gage
127. Depth of cartridge head recess including chamfer	Gage

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4.3.2.3 Cap, Initiator, Dwg. D10520473 Covering a Detail of Drawing D10520909 - Cont'd

	Defect	Method of Inspection
MAJOR		
128.	Length of effective exterior thread, min	Gage
129.	Length of end of effective exterior thread, max	Gage
130.	Length of effective small interior thread, min	Gage
131.	Thickness of boss, min	Gage
132.	Location of pin hole	Gage
133.	Chamfer at striker hole missing or incorrect	Visual
134.	Protective finish inadequate or defective	Visual
135.	Surface finish of counterbore, mouth and small cavity improper	Visual
136.	Burr or sharp corner	Visual
137.	Foreign matter	Visual
MINOR		
201.	Pitch diameter of large internal thread, min	Gage
202.	Pitch diameter of small internal thread, min	Gage
203.	Pitch diameter of external thread, max	Gage
204.	Pitch diameter of pin hole thread, min	Gage
205.	Pitch diameter of set screw thread	Gage
206.	Minor diameter of set screw thread, max	Gage
207.	Diameter of boss, max	Gage
208.	Diameter of striker hole, max	Gage
209.	Diameter of pin hole, max	Gage
210.	Diameter of body, large end	Gage
211.	Depth to bottom of counterbore, small threaded end, min	Gage
212.	Depth of small threaded cavity, including chamfer, max	Gage
213.	Depth of large cavity including undercut	Gage
214.	Total length	Gage
215.	Length of body, large end	Gage
216.	Length from face of large cavity to shoulder	Gage
217.	Length of flat	Gage
218.	Width across flats	Gage
219.	Width of large thread undercut	Gage
220.	Thickness of boss, max	Gage

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4.3.2.3 - Cont'd

Defect	Method of Inspection
MINOR	
221. Location of set screw hole	Gage
222. Radius or chamfer missing or incorrect	Visual
223. Surface finish improper except as otherwise classified	Visual

4.3.2.4 Pin, Firing C10520474, Covering a Detail of Drawing D10520909

Defect	Method of Inspection
CRITICAL	
1. Metal defective	Visual
MAJOR	
101. Concentricity of striker with body	Gage
102. Concentricity of groove with body	Gage
103. Symmetry of pin hole with body	Gage
104. Diameter of striker	Gage
105. Diameter of body	Gage
106. Diameter of groove	Gage
107. Diameter of pin hole	Gage
108. Depth of pin hole	Gage
109. Length of striker, min	Gage
110. Location of pin hole	Gage
111. Protective finish inadequate or defective	Visual
112. Surface finish of body improper	Visual
113. Burr or charp corner	Visual
114. Foreign matter	Visual
MINOR	
201. Total length	Gage
202. Length of striker, max	Gage
203. Location of groove	Gage
204. Depth of slot	Gage
205. Width of slot	Gage
206. Width of groove	Gage
207. Location of slot	Gage
208. Angle on striker end of body	Gage
209. Radius or chamfer missing or incorrect	Visual
210. Surface finish improper, except as otherwise classified	Visual

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4.3.2.5 Pin, Shear, Drawing C10521653, Covering a Detail of Drawing D10520909

Defect	Method of Inspection
CRITICAL	
None defined	
MAJOR	
101. Diameter	Gage
102. Total length	Gage
103. Metal defective	Visual
104. Protective finish inadequate or defective	Visual
105. Burr	Visual
106. Foreign matter	Visual
MINOR	
201. Length of chamfer on end	Gage
202. Surface finish improper	Visual

4.3.2.6 Assembly, Drawing D10520909, Less all Drawings except D10520473, C10520474 and MS29513-9.

CRITICAL	
1. Firing pin protrusion	Gage
2. "O" ring missing	Visual
MAJOR	
101. Burr	Visual
102. Foreign matter	Visual
MINOR	
None defined	

4.3.2.7 Assembly, Drawing D10520909

CRITICAL	
1. Any part missing except as otherwise classified	X-Ray and Visual

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4.3.2.7 Assembly, Drawing D10520909 - Cont'd

Defect	Method of Inspection
CRITICAL	
2. Improper assembly	X-Ray 1/ and Visual
3. Metal defective	X-Ray 1/ and Visual
4. Marking missing, incorrect or illegible	Visual
MAJOR	
101. Thread damaged or not full	Visual
102. "O" ring not lubricated	Visual
MINOR	
201. Shipping cap or plug missing	Visual
202. Moisture and fingers resistant varnish missing or inadequate	Visual
203. Protective finish inadequate or defective	Visual

1/ The initiator shall be x-rayed with the sides of the case perpendicular to the rays. The x-ray negatives properly identified shall be forwarded to the procuring activity.

4.3.2.8 Fiber Container, Drawing C8846787, Container, Ammunition, Fiber M458 for Initiators Cartridge Actuated Delay

Defect	Method of Inspection
CRITICAL	
None defined	
MAJOR	
None defined	
MINOR	
201. Contents loose	Manual
202. Metal ends loose or distorted	Manual/Visual
203. Gap between body and cover, max	Visual
204. Cut or damaged through all impregnated layers	Visual

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4.3.2.9 Box, Packing, Ammunition for Initiators, Cartridge Actuated, Delay in Fiber Container M458

Defect		Method of Inspection
CRITICAL		
	None defined	
MAJOR		
	None defined	
MINOR		
201.	Content loose	Manual
202.	Hardware missing, loose, broken or improperly engaged	Manual and Visual
203.	Strapping missing, loose, broken or improperly assembled	Manual and Visual
204.	Content of box exposed	Visual
205.	Box damaged	Visual
206.	Marking missing, incorrect or illegible	Visual
207.	Number of unit incorrect	Visual
208.	Car seal loose, missing or improperly positioned	Visual

4.3.3 Inspection testing - The following test shall be performed in accordance with the provisions of 4.4.

4.3.3.1 Hardness - Fifty units shall be selected for this test. Failure of any unit to comply with the requirement of 3.2 shall be cause for rejection of the lot.

4.3.3.2 Hydrostatic pressure - Each assembly shall be subjected to the hydrostatic pressure test. Failure of any unit to comply with the requirement of 3.3 shall be cause for rejection of the assembly.

4.3.3.3 Shear pin strength - Fifty pins from each lot shall be tested for shear strength. Failure of any shear pin to comply with the requirement of 3.4 shall be cause for rejection of the lot.

4.3.3.4 Protective finish - The number of unit for protective finish tests shall be in accordance with applicable specification. Failure of any sample to comply with the requirement of the applicable specification shall be cause for rejection of the lot.

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4.3.3.5 Residual magnetism - Each assembly shall be subjected to the test for excessive residual magnetism. Failure of any initiator to comply with the requirement of 3.6 shall be cause for rejection of the assembly.

4.3.4 Inspection Equipment - The examination and tests shall be made using equipment listed on IEL 10520909.

4.4 Test Methods and Procedures

4.4.1 Hardness - Prior to the application of protective finish, a hardness testing machine shall be used to determine compliance with hardness requirements specified on the applicable drawing.

4.4.2 Hydrostatic Pressure - Prior to the application of protective finish, the chamber and cap sub-assembly shall be mounted in the device using the fixture for properly sealing all openings and subjected to the hydrostatic test to determine compliance with 3.3.

4.4.3 Shear pin strength - After application of protective finish, the pin shall be subjected to the shear pin strength test to determine compliance with the requirement of 3.4. The test shall be performed in the fixture in connection with a tensile testing machine.

4.4.4 Protective finish - Test procedures for protective finish shall be in accordance with the specification referenced on the component drawing.

4.4.5 Residual magnetism - The initiator shall be tested for excessive residual magnetism in an area free of local magnetic effects by placing the initiator 6 inches from and at the same height as the compass, in a north-south horizontal position (cap end at south) with the compass set in an east-west heading. The initiator shall be moved slowly past the compass in a longitudinal direction for its entire length, rotated 90 degrees in the horizontal position about its longitudinal axis and moved slowly back past the compass. This procedure shall be repeated for each 90 degree rotation of the initiator.

5. PREPARATION FOR DELIVERY

5.1 Packing and marking - Packing and marking shall be in accordance with drawings C8846787 and C8846788.

6. NOTES

6.1 Ordering data - Procurement document should specify the title, number and date of this specification.

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6.2 A mechanical failure is defined as any deformation or breakage of a part, the occurrence of which is other than a design function.

6.3 Supersession data - This specification includes the requirements of Purchase Description FA-PD-MI-2428 dated 26 April 1963.

Custodian

Army - MU

Preparing Activity:

Army - MU - FA

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SPECIFICATION ANALYSIS SHEET

Form Approved
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INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

[ ] DIRECT GOVERNMENT CONTRACT [ ] SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

[ ] YES [ ] NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

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REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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